

Anesthesia for Obese Patients (Bariatric Anesthesia) MCQ – Chapter 31

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 31)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What is the definition of obesity based on BMI?

- A. BMI more than 20%
- B. BMI more than 25%
- C. BMI more than 30%
- D. BMI more than 40%

Ans: C

Q2. What BMI defines extreme/superobese?

- A. More than 60%
- B. More than 50%
- C. More than 30%
- D. More than 40%

Ans: B

Q3. Why are obese patients one of the most challenging for anesthesiologists?

- A. Due to positioning only
- B. Due to long procedures only
- C. Due to associated comorbidities and difficult airway management
- D. Due to cost issues

Ans: C

Q4. What cardiovascular conditions may obese patients have?

- A. Only hypotension
- B. Only hypertension
- C. Only bradycardia
- D. Hypertension, coronary artery disease, LVH, RVH, and cor pulmonale

Ans: D

Q5. How much does cardiac output increase per kg of fat to perfuse extra fat?

- A. 0.5 L/min
- B. 0.1 L/min
- C. 0.05 L/min
- D. 0.2 L/min

Ans: B

Q6. Why do obese patients develop right ventricular hypertrophy?

- A. Due to mitral regurgitation
- B. Due to fluid overload
- C. Due to aortic stenosis
- D. Resulting from pulmonary hypertension which is because of persistent hypoxia

Ans: D

Q7. What causes cor pulmonale in obese patients?

- A. Left ventricular failure
- B. Aortic stenosis
- C. Mitral stenosis
- D. Right ventricular strain

Ans: D

Q8. What respiratory changes are seen in obese patients? A. Increased FRC B. Reduced ERV, FRC, vital capacity, increased airway resistance, and reduced oxygen saturation C. Normal lung function D. Increased vital capacity	Ans: B
Q9. What is obstructive sleep apnea hypopnea syndrome (OSAHS) and its significance? A. Patients with OSAHS are at increased risk of morbidity and mortality B. It has no anesthetic significance C. A benign condition D. It only causes snoring	Ans: A
Q10. What GI complications may obese patients have? A. Only liver disease B. Only diarrhea C. Only constipation D. Hiatus hernia, gastroesophageal reflux, delayed gastric emptying and fatty infiltration of liver	Ans: D
Q11. Why may obese patients have polycythemia? A. Due to dehydration B. Due to EPO injections C. Due to iron overload D. Because of chronic hypoxia	Ans: D
Q12. What are obese patients very prone to in postoperative period? A. Thromboembolism B. Hyperthermia C. Bradycardia D. Hypothermia	Ans: A
Q13. Why do obese patients often have difficult airway? A. Short neck (decreased thyromental distance), restricted mouth opening, and restricted neck movement B. Due to dental issues C. Due to large tongue only D. Due to small mouth only	Ans: A
Q14. Is obesity per se the criteria to order routine tests? A. No, investigation should be based on clinical assessment and associated comorbidities B. Only blood tests are needed C. Yes, all tests must be done D. Only chest X-ray is needed	Ans: A

Q15. What must be checked by pulse oximetry as baseline in obese patients? A. Temperature B. Heart rate C. Oxygen saturation D. Blood pressure	Ans: C
Q16. In which obese patients should polysomnography be strongly considered? A. Morbidly obese patients or patients with a history of obstructive sleep apnea B. All obese patients C. Only those with hypertension D. Only super-obese patients	Ans: A
Q17. Why should respiratory depressant drugs like benzodiazepines be avoided for premedication in obese patients? A. Due to risk of respiratory depression in patients already prone to hypoxia and sleep apnea B. They cause nausea C. They are expensive D. They cause hypertension	Ans: A
Q18. What premedication should be considered for obese patients? A. Benzodiazepines B. Antihistamines only C. Opioids D. Metoclopramide and ranitidine	Ans: D
Q19. What size BP cuff should be used for monitoring obese patients? A. Appropriate size BP cuff B. Standard adult cuff C. Pediatric cuff D. Any available cuff	Ans: A
Q20. What is the anesthesia of choice for obese patients due to difficult airway? A. Regional anesthesia B. IV sedation C. General anesthesia D. Local anesthesia only	Ans: A
Q21. Why is regional anesthesia often difficult in obese patients? A. Due to infection risk B. Due to bleeding risk C. Due to reduced epidural and subarachnoid space making them vulnerable for high spinals D. Due to patient refusal	Ans: C

Q22. What has been found beneficial during preoxygenation in obese patients? A. Nebulization B. High flow nasal oxygen C. Adding 10 cm H₂O of CPAP during preoxygenation D. Incentive spirometry	Ans: C
Q23. Why do lipid soluble drugs (IV induction agents, opioids) have increased volume of distribution in obese patients? A. Due to increased renal clearance B. Due to increased fat stores C. Due to hepatic enzyme induction D. Due to decreased protein binding	Ans: B
Q24. On what weight basis should lipid soluble drugs be given in obese patients? A. Adjusted body weight B. Actual body weight C. Lean body weight D. Ideal body weight	Ans: B
Q25. Is the increased volume of distribution effect as marked on propofol as on thiopentone? A. More marked on propofol B. No, this effect is not so marked on propofol as on thiopentone C. Neither is affected D. Yes, equally marked	Ans: B
Q26. What can achieving IV lines in obese patients become? A. Impossible always B. An arduous task C. A simple procedure D. Very easy	Ans: B
Q27. What is the most challenging part in obese patients? A. Positioning B. Monitoring C. Airway management D. Drug dosing	Ans: C
Q28. How much more is the incidence of failed intubation in bariatric population? A. 5 times more B. 10 times more C. 3 times more D. 2 times more	Ans: C

Q29. What is the preferred position for intubation in obese patients? A. Ramped position (ear to sternum position) B. Trendelenburg position C. Lateral position D. Supine position	Ans: A
Q30. How is the ramp position achieved? A. By flexing the neck only B. By extending the head only C. By tilting the table D. By elevating the upper back using sheets/blankets so external auditory meatus lies in horizontal line with sternal notch	Ans: D
Q31. What should be considered if intubation is anticipated to be difficult in obese patients? A. Tracheostomy first B. LMA only C. Awake intubation with fiberoptic bronchoscopy D. Blind nasal intubation	Ans: C
Q32. How must confirmation of intubation be done in obese patients? A. By capnography as breath sounds may not be clearly audible on auscultation B. By auscultation only C. By tube length marking D. By chest rise only	Ans: A
Q33. Why are inhalational agents more extensively metabolized in obese patients? A. Due to increased fat stores and hepatic blood flow B. Due to decreased protein binding C. Due to increased renal clearance D. Due to enzyme deficiency	Ans: A
Q34. Which is the inhalational agent of choice for maintenance in obese patients? A. Halothane B. Isoflurane C. Sevoflurane D. Desflurane	Ans: D
Q35. Why should halothane be avoided in obese patients? A. Increased metabolism of halothane makes obese patients more vulnerable for halothane hepatitis B. It causes arrhythmias C. It is not potent enough D. It causes bronchospasm	Ans: A

Q36. On what weight basis should water soluble drugs (nondepolarizing muscle relaxants) be given? A. Actual body weight B. Ideal body weight C. Total body weight D. Adjusted body weight	Ans: B
Q37. What does application of 10 cm H₂O PEEP during mechanical ventilation do in obese patients? A. Decreases oxygenation B. Improves oxygenation by preventing alveolar collapse C. Causes hypotension D. Has no effect	Ans: B
Q38. What position improves oxygenation particularly during laparoscopic surgeries in obese patients? A. Lateral decubitus B. Supine flat C. 30 degree reverse Trendelenburg D. Trendelenburg	Ans: C
Q39. What are obese patients very prone to in the postoperative period? A. Hypothermia B. Hypoxia C. Bradycardia D. Hyperglycemia only	Ans: B
Q40. In what position should obese patients be nursed postoperatively? A. Trendelenburg B. Supine flat C. 45 degree sitting (semi-reclined position) with supplemental oxygen D. Prone	Ans: C
Q41. When should CPAP or BiPAP be applied in postoperative obese patients? A. Only after 24 hours B. If patient is not maintaining oxygen saturation on supplemental oxygen C. Never needed D. Always routinely	Ans: B
Q42. What must be taken to prevent thromboembolism postoperatively in obese patients? A. Every precaution (DVT prophylaxis) B. Only aspirin C. Only compression stockings D. Only bed rest	Ans: A

Q43. Why is good antibiotic cover a must in obese patients? A. Due to poor hygiene only B. Due to diabetes only C. Due to immunosuppression D. Because these patients are prone for wound infection	Ans: D
Q44. Why are obese patients in high cardiac output state? A. To perfuse extra fat tissue B. Due to fever C. Due to anxiety D. Due to exercise	Ans: A
Q45. What restrictive pattern do obese patients have in their pulmonary functions? A. Mixed pattern B. Normal pattern C. Extra parenchymal restrictive pattern D. Obstructive pattern	Ans: C
Q46. What does the ramped position provide for laryngoscopy? A. No improvement B. Superior view C. Lateral view D. Inferior view	Ans: B
Q47. What improves oxygenation in obese patients through application of CPAP and PEEP? A. Decreasing dead space B. Decreasing respiratory rate C. Preventing alveolar collapse and improving functional residual capacity D. Increasing tidal volume	Ans: C
Q48. Why is decreased thyromental distance a concern in obese patients? A. It indicates thyroid disease B. It indicates short neck making intubation difficult C. It indicates dental problems D. It has no significance	Ans: B
Q49. What causes reduced oxygen saturation in obese patients? A. Pulmonary fibrosis B. Anemia C. Obesity itself and obstructive sleep apnea D. Heart failure only	Ans: C

Q50. Why is the ramp position also called ear to sternum position?

- A. Because neck is hyperextended
- B. Because ear touches the sternum
- C. Because external auditory meatus lies in horizontal line with sternal notch
- D. Because head is below sternum

Ans: C

Answer Key & Explanations

Q1. Answer: C — BMI more than 30%

Obesity is defined as body mass index (BMI) more than 30% and extreme/superobese as BMI more than 50%.

Q2. Answer: B — More than 50%

Extreme/superobese is defined as BMI more than 50%.

Q3. Answer: C — Due to associated comorbidities and difficult airway management

Obese patients are one of the most challenging patients for anesthesiologist especially due to their associated comorbidities and difficult airway management.

Q4. Answer: D — Hypertension, coronary artery disease, LVH, RVH, and cor pulmonale

Obese patients may have hypertension, coronary artery disease, left ventricular hypertrophy, right ventricular hypertrophy from pulmonary hypertension, and cor pulmonale.

Q5. Answer: B — 0.1 L/min

Increase in cardiac output by 0.1 L/min to perfuse each kg of fat leads to increased stroke volume and hence left ventricular hypertrophy.

Q6. Answer: D — Resulting from pulmonary hypertension which is because of persistent hypoxia

Right ventricular hypertrophy resulting from pulmonary hypertension which in turn is because of persistent hypoxia.

Q7. Answer: D — Right ventricular strain

Cor pulmonale because of right ventricular strain.

Q8. Answer: B — Reduced ERV, FRC, vital capacity, increased airway resistance, and reduced oxygen saturation

Obese patients may have reduced expiratory reserve volume (ERV), functional residual capacity (FRC) and vital capacity, increased airway resistance, and reduced oxygen saturation.

Q9. Answer: A — Patients with OSAHS are at increased risk of morbidity and mortality

Patients with OSAHS are at increased risk of morbidity and mortality.

Q10. Answer: D — Hiatus hernia, gastroesophageal reflux, delayed gastric emptying and fatty infiltration of liver

Obese patients may have associated hiatus hernia, gastroesophageal reflux, delayed gastric emptying and fatty infiltration of liver (may compromise liver functions).

Q11. Answer: D — Because of chronic hypoxia

There may be polycythemia because of chronic hypoxia.

Q12. Answer: A — Thromboembolism

Obese patients are very prone for thromboembolism in postoperative period.

Q13. Answer: A — Short neck (decreased thyromental distance), restricted mouth opening, and restricted neck movement

Obese patients often have difficult airway due to short neck (decreased thyromental distance), restricted mouth opening due to increased adipose tissue and decreased mobility of temporomandibular joint, and restricted neck movement because of decreased mobility of atlanto-occipital joint.

Q14. Answer: A — No, investigation should be based on clinical assessment and associated comorbidities

Obesity per se is not the criteria to order a battery of routine tests, therefore investigation should be based on clinical assessment and associated comorbidities.

Q15. Answer: C — Oxygen saturation

Baseline oxygen saturation must be checked by pulse oximetry (if abnormal, then arterial blood gases should be done).

Q16. Answer: A — Morbidly obese patients or patients with a history of obstructive sleep apnea

Polysomnography should be strongly considered in morbidly obese patients or in patients with a history of obstructive sleep apnea.

Q17. Answer: A — Due to risk of respiratory depression in patients already prone to hypoxia and sleep apnea

Premedication with respiratory depressant drugs like benzodiazepines should be avoided.

Q18. Answer: D — Metoclopramide and ranitidine

Premedication with metoclopramide and ranitidine should be considered (due to hiatus hernia and GERD risk).

Q19. Answer: A — Appropriate size BP cuff

Appropriate size BP cuff should be chosen for monitoring.

Q20. Answer: A — Regional anesthesia

Due to difficult airway management, anesthesia of choice is regional anesthesia.

Q21. Answer: C — Due to reduced epidural and subarachnoid space making them vulnerable for high spinals

Unfortunately regional anesthesia is often difficult in obese patients. Due to reduced epidural and subarachnoid space obese patients are more vulnerable for high spinals.

Q22. Answer: C — Adding 10 cm H₂O of CPAP during preoxygenation

Adding 10 cm H₂O of continuous positive airway pressure (CPAP) during preoxygenation has found to be beneficial in maintaining oxygenation in the perioperative period.

Q23. Answer: B — Due to increased fat stores

Lipid soluble drugs (IV induction agents, opioids) have increased volume of distribution, therefore given on actual body weight basis.

Q24. Answer: B — Actual body weight

Lipid soluble drugs have increased volume of distribution, therefore given on actual body weight basis.

Q25. Answer: B — No, this effect is not so marked on propofol as on thiopentone

Interestingly, this effect is not so marked on propofol as on thiopentone.

Q26. Answer: B — An arduous task

Achieving IV lines sometimes can become an arduous task.

Q27. Answer: C — Airway management

Airway management is the most challenging part in obese patients.

Q28. Answer: C — 3 times more

The incidence of failed intubation is 3 times more in bariatric population.

Q29. Answer: A — Ramped position (ear to sternum position)

The Ramped position provides a superior view of laryngoscopy and is the preferred position for intubation for obese patients.

Q30. Answer: D — By elevating the upper back using sheets/blankets so external auditory meatus lies in horizontal line with sternal notch

A ramp is created by using sheets and blankets to elevate the upper back of the patient to the level when external auditory meatus lie in horizontal line with the sternal notch (ear to sternum position).

Q31. Answer: C — Awake intubation with fiberoptic bronchoscopy

If intubation is anticipated to be difficult awake intubation with fiberoptic bronchoscopy should be considered.

Q32. Answer: A — By capnography as breath sounds may not be clearly audible on auscultation

Confirmation of intubation must be done by capnography as breath sounds may not be clearly audible on auscultation.

Q33. Answer: A — Due to increased fat stores and hepatic blood flow

Inhalational agents are more extensively metabolized in obese patients.

Q34. Answer: D — Desflurane

Desflurane is the inhalational agent of choice because it is the agent with least metabolism.

Q35. Answer: A — Increased metabolism of halothane makes obese patients more vulnerable for halothane hepatitis

Increased metabolism of halothane makes obese patients more vulnerable for halothane hepatitis.

Q36. Answer: B — Ideal body weight

Water soluble drugs (nondepolarizing muscle relaxants) have limited volume of distribution therefore should be given on ideal body weight basis.

Q37. Answer: B — Improves oxygenation by preventing alveolar collapse

Application of 10 cm H₂O of positive end expiratory pressure (PEEP) during mechanical ventilation improves oxygenation by preventing alveolar collapse.

Q38. Answer: C — 30 degree reverse Trendelenburg

30 degree reverse Trendelenburg improves oxygenation particularly during laparoscopic surgeries.

Q39. Answer: B — Hypoxia

Obese patients are very prone to hypoxia in postoperative period.

Q40. Answer: C — 45 degree sitting (semi-reclined position) with supplemental oxygen

They should be nursed in 45 degree sitting (semi-reclined position) with supplemental oxygen.

Q41. Answer: B — If patient is not maintaining oxygen saturation on supplemental oxygen

CPAP or BiPAP should be applied if the patient is not maintaining oxygen saturation on supplemental oxygen.

Q42. Answer: A — Every precaution (DVT prophylaxis)

Every precaution must be taken to prevent thromboembolism.

Q43. Answer: D — Because these patients are prone for wound infection

Good antibiotic cover is must as these patients are prone for wound infection.

Q44. Answer: A — To perfuse extra fat tissue

To perfuse extra fat obese patients are in high cardiac output state.

Q45. Answer: C — Extra parenchymal restrictive pattern

Obese patients may have extra parenchymal restrictive pattern in their pulmonary functions.

Q46. Answer: B — Superior view

The ramped position provides a superior view of laryngoscopy.

Q47. Answer: C — Preventing alveolar collapse and improving functional residual capacity

Application of CPAP and PEEP improves oxygenation in obese patients by preventing alveolar collapse.

Q48. Answer: B — It indicates short neck making intubation difficult

Short neck (decreased thyromental distance) contributes to difficult airway in obese patients.

Q49. Answer: C — Obesity itself and obstructive sleep apnea

Reduced oxygen saturation because of obesity and obstructive sleep apnea.

Q50. Answer: C — Because external auditory meatus lies in horizontal line with sternal notch

It is also called ear to sternum position because external auditory meatus lie in horizontal line with the sternal notch.